Biology Extended Essay planner

What the Cell · DP Biology · extended essay. Write your school's deadlines next to each stage. The online version is at whatthecell.org/extended-essay.html

What it is	An independent research essay of up to 4,000 words on a question you choose, here in Biology
Time	About 40 hours of your own work, spread over roughly a year
Marks	5 criteria, 30 marks in total, marked by an IB examiner
Reflection	Three meetings with your supervisor, then a reflection statement of up to 500 words
Why it matters	Together with TOK it can add up to 3 points to your Diploma, and you need at least a D to pass the Diploma

These facts follow the IB Extended Essay guide for essays first assessed in May 2027 (30 marks, five criteria). Your school sets the exact deadlines, so copy them into the planner below.

EE or IA? Your IA answers one focused question in 3,000 words with about 10 hours of work. Your EE goes deeper: 4,000 words, about 40 hours, more background reading, and a fuller discussion of what your results mean. The same work cannot be used for both.

The five criteria in plain words

Criterion	Marks	What the examiner wants to see	Where you build it
A. Framework for the essay	6	A sharp research question, a method that suits it, and a clear structure with headings, labelled tables and graphs, and consistent referencing	Stages 2, 3 and 8
B. Knowledge and understanding	6	Biology that is correct, relevant and explained in the right terms, supported by good sources	Stages 2, 4 and 7
C. Analysis and line of argument	6	Data or evidence that is analysed properly, and an argument that runs from the question to the conclusion without losing its way	Stages 6 and 7
D. Discussion and evaluation	8	What your findings mean, how far they answer the question, how reliable they are, and how they fit with published work	Stages 7 and 8
E. Reflection	4	An honest account of how your thinking and skills changed during the research, in your reflection statement	Stages 1 to 9

Criterion D carries the most marks. Leave real time for it: most weak EEs stop at the results.

The staged plan

The plan assumes a school year from August to June, with the essay finished early in the second year. Ask your coordinator for the real dates.

Stage	When (typical)	What you do	Done when
1. Choose Biology and a supervisor	Year 1, February	Check that you are taking Biology (strongly advised). Read three or four strong Biology EEs. Meet your supervisor	You have a supervisor and a shared folder
2. Explore a topic	Year 1, March	Read widely around two or three topics. Note what is known, what is argued about and what you could measure	You can explain each topic in two minutes
3. Research question and first reflection session	Year 1, April	Narrow to one focused, answerable question. Meet your supervisor for the first formal reflection session	Your question is approved
4. Plan, safety and ethics	Year 1, May	Write a research plan: approach, variables, method, materials, risk assessment and ethics check. Find the key sources	Your supervisor signs off the plan
5. Pilot and data collection	Year 1, May to August	Run a small pilot, fix problems, then collect your full data. Record everything in your research log	You have all your raw data
6. Analysis	Year 2, August to September	Process the data, draw the graphs, run the right statistical tests and treat uncertainties	Every result is in a table or graph
7. Interim reflection session and full draft	Year 2, September to October	Meet your supervisor for the interim session. Write the complete essay: introduction, background, method, results, discussion, conclusion and evaluation	A complete draft goes to your supervisor
8. Feedback and final version	Year 2, November	Your supervisor gives feedback on one complete draft. Revise, check every reference, the word count and the layout	Final essay handed in
9. Final session (viva voce) and reflection statement	Year 2, December	A short final interview with your supervisor. Then write your reflection statement (up to 500 words) on the reflection form	Statement handed in

Checklist

- [ ] My question names the organism, the variables or system, and what I will measure.
- [ ] I can get enough data in the time I have, with equipment the school has.
- [ ] My plan has been checked for safety, ethics and environmental impact.
- [ ] I keep a research log: dates, decisions, problems, sources and photos.
- [ ] Every source is recorded as I read it.
- [ ] I have used and declared any AI tools in line with school rules.

- [ ] My draft has a discussion and an evaluation, not only results.
- [ ] My final word count is 4,000 or fewer.
- [ ] My reflection statement is 500 words or fewer and written after the final session.

Choosing a Biology research question

A Biology EE needs biology at its heart: living things, their parts or their processes. Good questions are narrow enough to answer well in 4,000 words.

Too broad	Better
How does pollution affect plants?	What is the effect of copper sulfate concentration (0 to 100 mg dm^{-3}) on the growth rate of *Lemna minor* over 14 days?
Is exercise good for you?	How does the duration of moderate exercise affect recovery heart rate in 16 to 18 year olds? (with consent)
Do antibiotics work?	How does the concentration of garlic extract affect the zone of inhibition of *Escherichia coli* K-12?
What lives in mangroves?	How does distance from the shore affect the species diversity of crabs in a mangrove forest?
Why are coral reefs dying?	Is there a correlation between sea surface temperature and the frequency of coral bleaching at reef sites in Southeast Asia from 2000 to 2020? (database)

Approaches that work in Biology

- **Laboratory experiments:** one independent variable changed across at least five values, with repeats.
- **Fieldwork:** comparing sites, gradients or seasons, with proper sampling.
- **Databases:** analysing large published data sets, for example GBIF, NCBI or long-term climate and population records. You must still ask your own question and do your own analysis.
- **Combinations:** for example a lab experiment backed by a database comparison.

Essays that only summarise what other people have written rarely score well in Biology, because criteria C and D reward your own analysis and discussion.

Ethics and safety

- No experiments that cause pain, harm or distress to animals. Observing animals in their habitat without disturbing them is fine.
- Human participants must give informed consent (parental consent if under 16), can withdraw at any time, and stay anonymous.
- Follow your school's rules for microorganisms, chemicals and fieldwork. Leave field sites as you found them.

Structure of a Biology EE

Section	About how many words	Purpose
Title page and contents	(not counted)	Title, research question, subject, word count
Introduction	300	Why the question matters and what you will investigate
Background biology	700 to 900	The theory your discussion will need, with sources
Method	500 to 700	Variables, materials, procedure, safety and ethics
Results and analysis	600 to 800	Processed data, graphs and statistics (tables do not count towards the words)
Discussion	800 to 1,000	What the results show, explained with biology and compared with published studies
Conclusion and evaluation	500	A clear answer to the question, the limitations, how much they matter, and next steps
References and appendices	(not counted)	Every source in one consistent style; raw data if needed

What the top marks look like

- **A:** the question is precise and stays the same from title page to conclusion. Headings, numbered figures, units and references are consistent.
- **B:** the biology is accurate and goes beyond the textbook where needed, using journal articles and reliable sources.
- **C:** every claim is backed by your data or sources, and the statistics match the type of data.
- **D:** you explain what the results mean, admit what they cannot show, weigh how much each limitation matters, and compare with other studies.
- **E:** your reflection shows how your ideas and skills changed, with real examples of decisions, setbacks and what you learned from them.

The reflection statement

You meet your supervisor three times: early on, at the interim stage and at the end (the viva voce). After the final session you write one reflection statement of up to 500 words on the reflection form. Only the first 500 words are read, and a missing statement scores zero for criterion E.

Good statements are personal and specific. They describe:

- how you chose and changed your question, and why
- a problem you hit and how you solved it
- what you would do differently
- what you learned about doing research, and about yourself as a learner

Keep a research log from the start: it is the raw material for your statement.

Common mistakes

- A question so broad that the essay becomes a textbook summary.
- Starting data collection too late, especially with living organisms.
- Results with no statistics, or statistics that do not suit the data.
- A conclusion that repeats the results instead of explaining them.
- Writing the reflection statement from memory at the last minute.
- References added at the end, so some sources can no longer be found.

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